

# **CUSTOMER BEHAVIOR ANALYSIS**

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**DAB303 –Project 3**

# BUSINESS OBJECTIVE OF THE PROJECT

- **Understand and gain insights from an e-Commerce dataset, by performing various exploratory data analysis, data visualization, and data modelling tasks**
- **Focus specifically on Customer Behavior Analysis (notions, applications, etc.)**
- **Perform Advanced Data Science Analysis**



# ABOUT THE DATASET – CUSTOMER BEHAVIORS

**3,333 rows**

**20 columns**

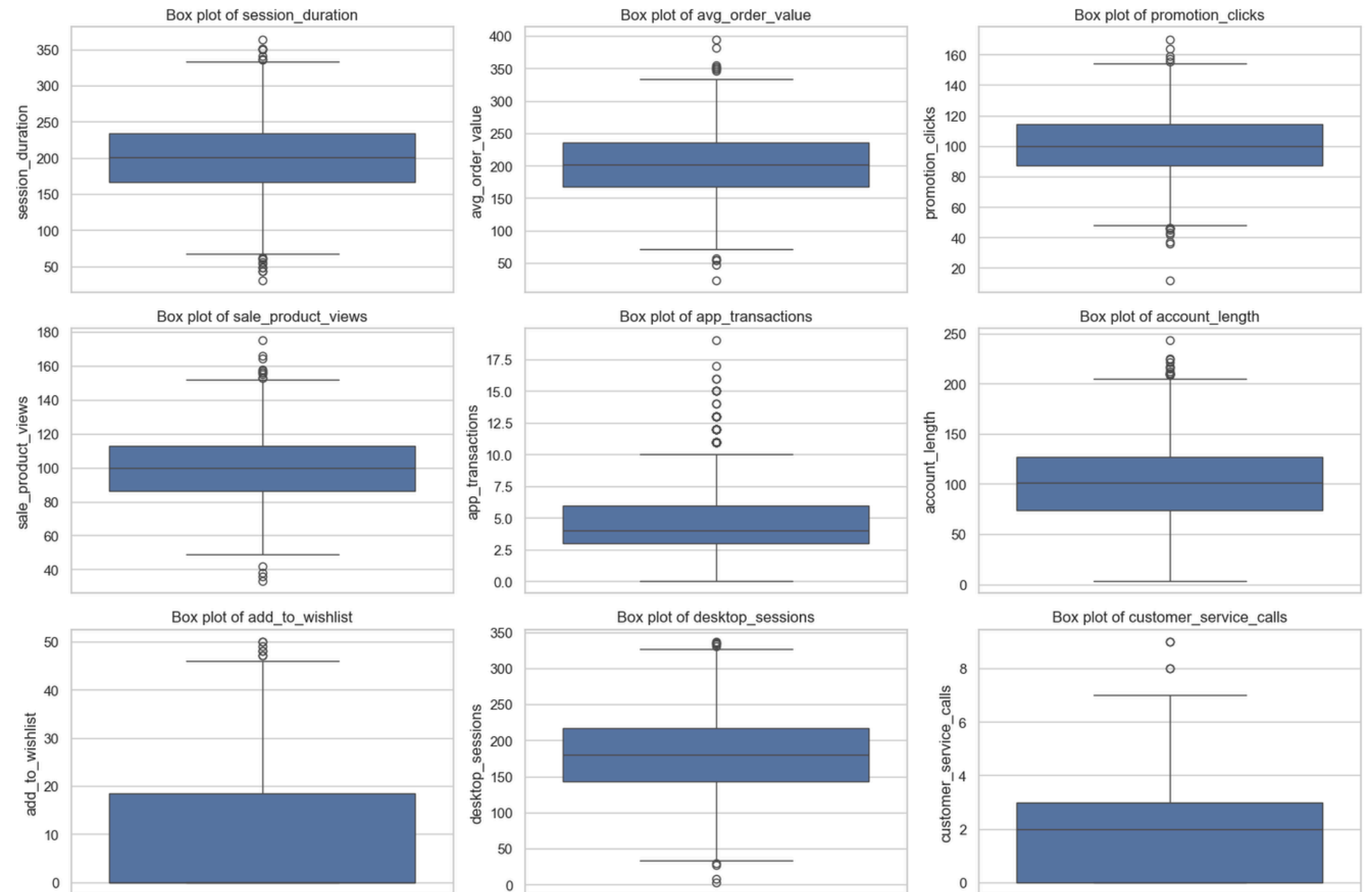
| Column                              | Description                                                                                      |
|-------------------------------------|--------------------------------------------------------------------------------------------------|
| account length                      | The duration (in days or months) since the user created their account.                           |
| location code                       | A code representing the geographic location of the user.                                         |
| user id                             | A unique identifier for each user.                                                               |
| credit card info save               | Indicates whether the user has saved their credit card information (1 for yes, 0 for no).        |
| push status                         | Indicates whether push notifications are enabled for the user (e.g., "enabled" or "disabled").   |
| add to wishlist                     | The count of how many times the user added items to their wishlist.                              |
| desktop sessions                    | The number of sessions the user had on desktop devices.                                          |
| app sessions                        | The number of sessions the user had on the mobile app.                                           |
| desktop transactions                | The number of transactions completed through desktop devices.                                    |
| total product detail views          | Total number of product detail pages viewed by the user across all sessions.                     |
| session duration                    | Total time spent by the user across all sessions (e.g., in minutes).                             |
| promotion clicks                    | The number of times the user clicked on promotions or ads.                                       |
| avg order value                     | The average value of orders placed by the user.                                                  |
| sale product views                  | The number of views for products that were on sale.                                              |
| discount rate per visited products  | The average discount rate on products viewed by the user.                                        |
| product detail view per app session | The number of product detail views per session on the mobile app.                                |
| app transactions                    | The number of transactions completed via the mobile app.                                         |
| add to cart per session             | The average number of products added to the cart per session.                                    |
| customer service calls              | The number of times the user contacted customer service.                                         |
| churn                               | Indicates whether the user has stopped engaging with the platform (1 for churned, 0 for active). |



# CHECK FOR OUTLIERS

## Visualization Outliers – Box Plots

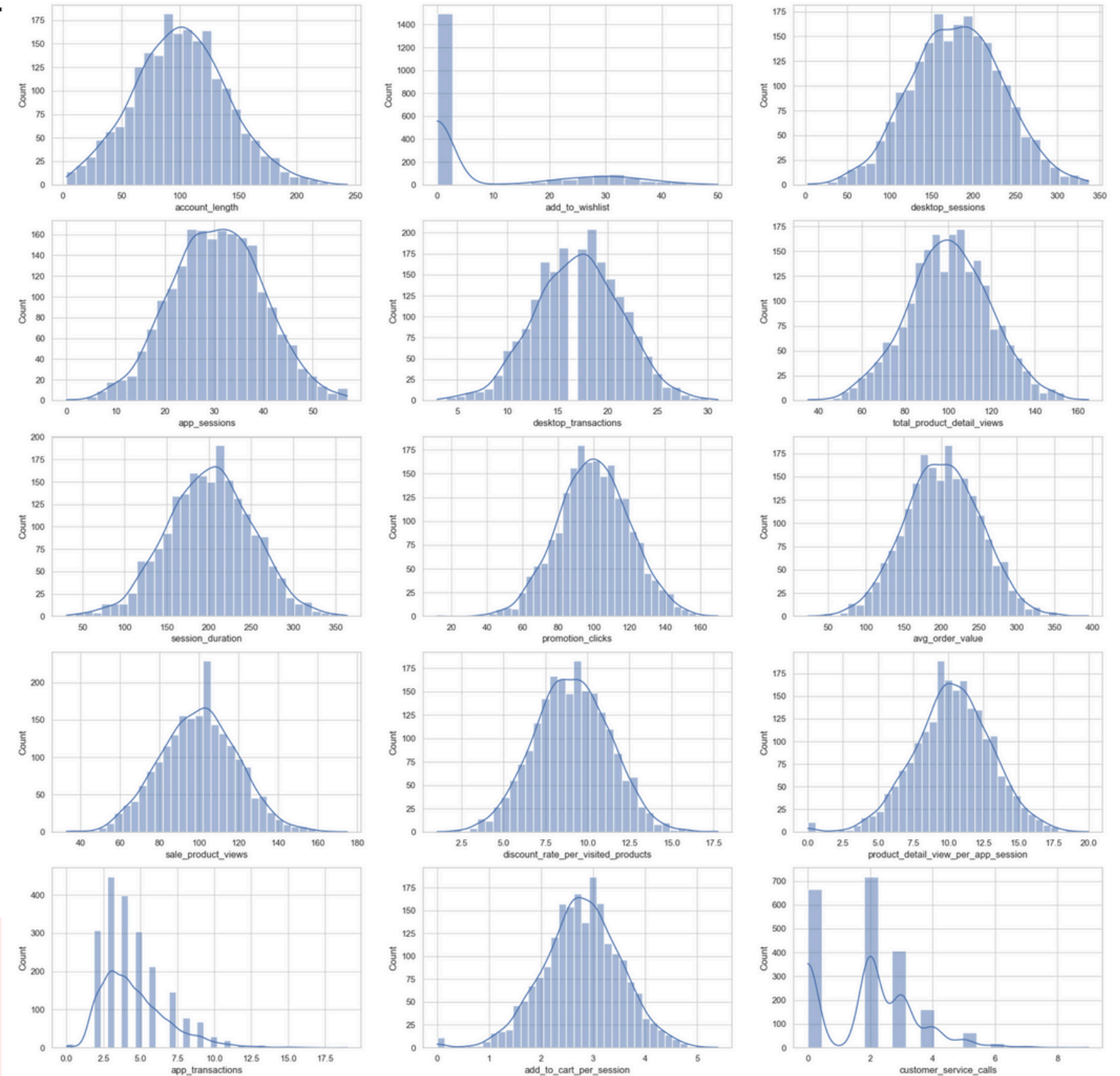
The analysis shows that most customer behaviors fall within a normal range, but several variables exhibit notable outliers. Session duration, average order value, and desktop sessions have typical mid-range activity with a few extremely high or low values. Most customers make around five app transactions, have accounts about 100 days old, and rarely contact customer service. Engagement levels, such as adding items to wishlists or clicking promotions, are generally low, though some users are highly active. Overall, the data suggests that while the majority of customers behave consistently, a small group shows significantly different interaction patterns.



# CHECK FOR OUTLIERS

## Visualization Outliers – Histograms

The histograms show that most features are normally distributed, such as **session duration**, **average order value**, **account length**, and **desktop sessions**, all centered around typical mid-range values. In contrast, app transactions, **add to wishlist**, and **customer service calls** are **right-skewed**, indicating that most users have low activity with a few highly active outliers. **Sale product views** and **promotion clicks** also show mild right skewness, reflecting that a majority of customers engage moderately while a smaller group is significantly more active.



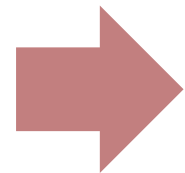
# CONCLUSION ABOUT VISUALIZATIONS

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- **The box plots** reveal the range and outliers in the data, showing that several features (like promotion clicks, app transactions, and add-to-wishlist events) have a significant number of outliers, which can be critical in further analysis.
- **The histograms** provide insights into the overall distribution of the data, with most features following a normal or right-skewed distribution, meaning that the majority of values are clustered around the median, with a tail extending to higher values.

## Observations:

- **Normality:** Session duration, average order value, promotion clicks, sale product views, and account length have distributions that resemble a normal distribution.
- **Skewness:** App transactions, add to wishlist, and customer service calls exhibit right-skewed distributions, indicating that most customers exhibit lower values for these metrics with few high-value outliers.
- **Outliers:** Box plots for app transactions, customer service calls, and add to wishlist highlight the presence of significant outliers, indicating users with unusual behavior.



Based on these 2 visualizations, it suggests that while most customer behaviors follow expected patterns, certain metrics show skewed or extreme behaviors that could warrant further investigation, especially regarding how outliers impact the overall business outcomes.

# ANALYSIS OF THE DEPENDENT VARIABLE (CHURN) VS INDEPENDENT VARIABLES

Until now we've looked into the numeric variables, their missing data, duplicates, distribution, and outliers. A key part of the customer behaviour data is to understand which "behaviours" aka independent variables in our dataset can help us predict customers deciding to stop using the product, aka "churn".

Another variable that we haven't analysed yet is the location code, but we might be able to find value in comparing it with the churn and finding out any potential correlations.

Features such as saving **credit\_card\_info\_save\_yes(0.2418)** and **customer\_service\_calls(0.2254)** show positive correlations with churn, indicating that customers engaging in these actions are more likely to churn.

On the other hand, features like **push\_status\_yes (-0.0887)** and **app\_transactions (-0.0774)** have negative correlations, suggesting that users receiving push notifications or conducting transactions in the app are less likely to churn.

## Correlation ships between 'churn' and others

|                                     |           |
|-------------------------------------|-----------|
| churn                               | 1.000000  |
| credit_card_info_save_yes           | 0.241808  |
| customer_service_calls              | 0.225397  |
| desktop_sessions                    | 0.169487  |
| app_sessions                        | 0.169275  |
| product_detail_view_per_app_session | 0.074796  |
| add_to_cart_per_session             | 0.074757  |
| session_duration                    | 0.070966  |
| desktop_transactions                | 0.068573  |
| avg_order_value                     | 0.024200  |
| discount_rate_per_visited_products  | 0.024192  |
| total_product_detail_views          | 0.023905  |
| user_id                             | 0.020291  |
| account_length                      | 0.018910  |
| location_code_510                   | 0.010762  |
| promotion_clicks                    | -0.001448 |
| sale_product_views                  | -0.009115 |
| location_code_415                   | -0.028283 |
| add_to_wishlist                     | -0.075246 |
| app_transactions                    | -0.077367 |
| push_status_yes                     | -0.088684 |

# PREDICTION OF CHURN

**Class 0 (Majority Class):**

- Precision: 0.94 – Out of all predicted class 0, 94% were correct.
- Recall: 0.99 – Out of all actual class 0, 99% were correctly classified.
- F1-score: 0.96 – The harmonic mean of precision and recall, showing excellent overall performance for class 0.
- Support: 340 – There were 340 instances of class 0.

**Class 1 (Minority Class):**

- Precision: 0.90 – Out of all predicted class 1, 90% were correct.
- Recall: 0.67 – Out of all actual class 1, only 67% were correctly classified, indicating the model is missing many of the actual class 1 instances.
- F1-score: 0.77 – A lower score compared to class 0, showing the classifier struggles with class 1.
- Support: 70 – There were 70 instances of class 1.

**Overall :**

- Accuracy: 0.93 – The model correctly classified 93% of the total instances, a good result overall.
  - Macro Average:
  - Precision: 0.92 – An unweighted average precision score.
  - Recall: 0.83 – The macro recall is lower, mainly due to the poor recall for class 1.
  - F1-score: 0.87 – Reflects the imbalanced performance between the two classes.
  - Weighted Average:
  - Precision: 0.93 – Accounts for the number of instances in each class.
  - Recall: 0.93 – Since class 0 has more instances, its high recall boosts the weighted average.
  - F1-score: 0.93 – Shows overall good performance but still reflects the imbalance.
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## Prediction using DecisionTreeClassifier training model

| Decision Tree Classifier: |           |        |          |         |  |
|---------------------------|-----------|--------|----------|---------|--|
|                           | precision | recall | f1-score | support |  |
| 0                         | 0.95      | 0.95   | 0.95     | 340     |  |
| 1                         | 0.77      | 0.76   | 0.76     | 70      |  |
| accuracy                  |           |        | 0.92     | 410     |  |
| macro avg                 | 0.86      | 0.86   | 0.86     | 410     |  |
| weighted avg              | 0.92      | 0.92   | 0.92     | 410     |  |

- => **Class 0:** The model performs very well on the majority class, with high precision (0.94) and recall (0.99), leading to a strong F1-score of 0.96.
- => **Class 1:** The model performs well in terms of precision (0.90), but its recall for class 1 is relatively low (0.67), indicating it struggles to identify all instances of class 1. This is also reflected in the lower F1-score (0.77).
- => **Overall:** The accuracy is high at 0.93, but the recall for class 1 (0.67) indicates a challenge in correctly identifying minority class instances. The macro average is lower than the weighted average due to this imbalance.

# PREDICTION OF CHURN

**Class 0 (Majority Class):**

- **Precision: 0.95** – Out of all predicted class 0, 95% were correctly classified.
- **Recall: 0.95** – Out of all actual class 0, 95% were correctly classified.
- **F1-score: 0.95** – The harmonic mean of precision and recall.
- **Support: 340** – There were 340 instances of class 0.

**Class 1 (Minority Class):**

- **Precision: 0.77** – Out of all predicted class 1, 77% were correct.
  - **Recall: 0.76** – Out of all actual class 1, 76% were correctly classified.
  - **F1-score: 0.76** – The harmonic mean of precision and recall.
  - **Support: 70** – There were 70 instances of class 1.
- **Accuracy: 0.92** – The model correctly classified 92% of the total instances.
  - **Macro Average: 0.86** for precision, recall, and F1-score – This is the unweighted average, treating both classes equally.
  - **Weighted Average: 0.92** – This average accounts for the number of instances in each class (class 0 has much more support than class 1).
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**Prediction using RandomForestClassifier  
training model**

| RandomForest Classifier: |           |        |          |         |  |
|--------------------------|-----------|--------|----------|---------|--|
|                          | precision | recall | f1-score | support |  |
| 0                        | 0.94      | 0.99   | 0.96     | 340     |  |
| 1                        | 0.90      | 0.67   | 0.77     | 70      |  |
| accuracy                 |           |        | 0.93     | 410     |  |
| macro avg                | 0.92      | 0.83   | 0.87     | 410     |  |
| weighted avg             | 0.93      | 0.93   | 0.93     | 410     |  |

The model performs better on the majority class (class 0) with high precision, recall, and F1-score. For the minority class (class 1), performance is lower, particularly in precision (0.77) and recall (0.76), indicating some difficulty in predicting the minority class.

The overall accuracy is quite high (0.92), but the difference in class performance is reflected in the macro and weighted averages.

# FEATURE IMPORTANCE FROM RANDOMFOREST

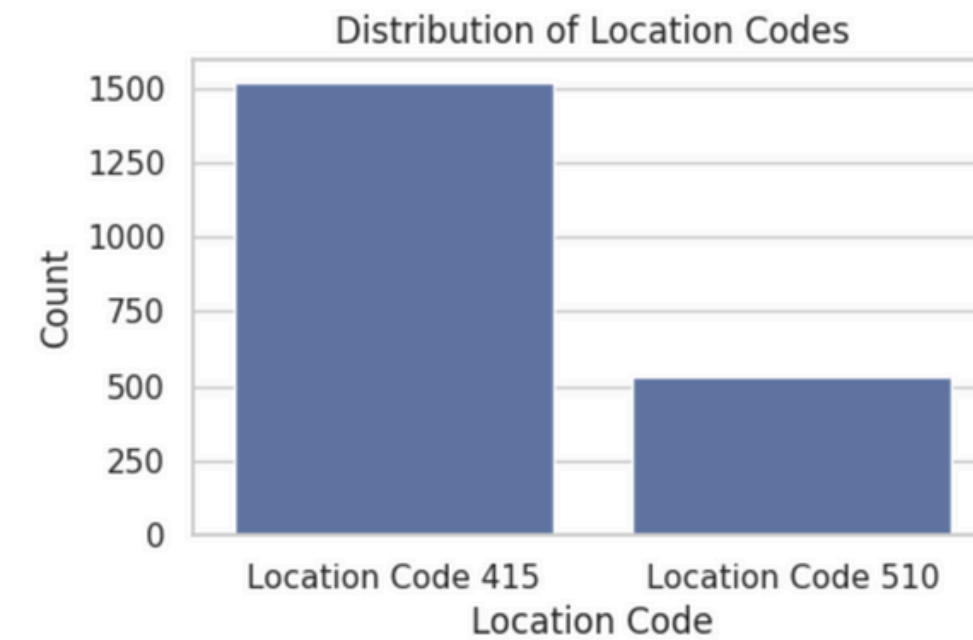
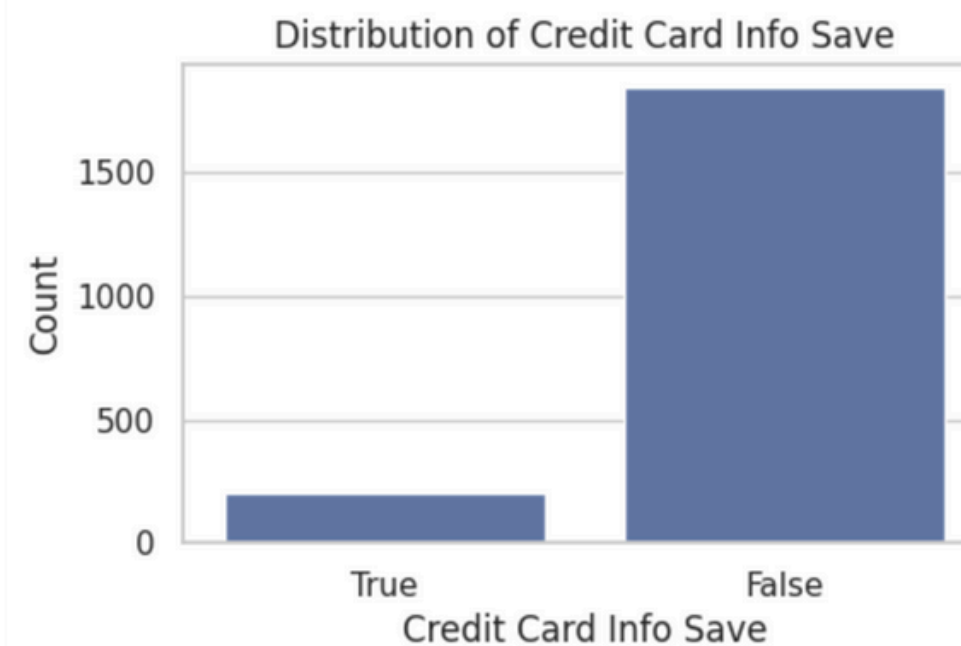
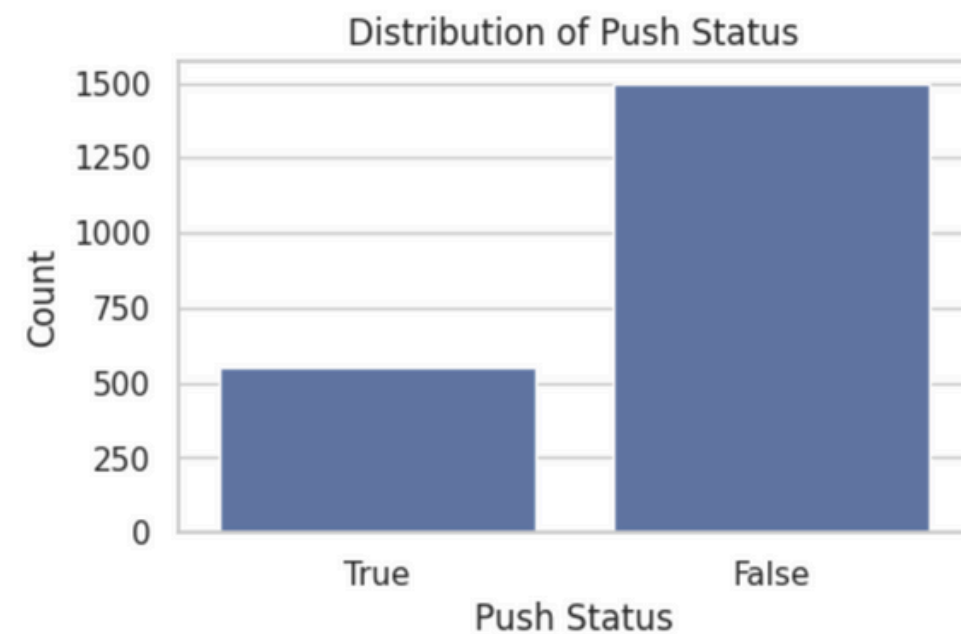
|    | Feature                             | Importance |
|----|-------------------------------------|------------|
| 3  | desktop_sessions                    | 0.155631   |
| 15 | customer_service_calls              | 0.138204   |
| 4  | app_sessions                        | 0.106852   |
| 7  | session_duration                    | 0.066338   |
| 16 | credit_card_info_save_yes           | 0.055305   |
| 5  | desktop_transactions                | 0.049162   |
| 14 | add_to_cart_per_session             | 0.043696   |
| 9  | avg_order_value                     | 0.039790   |
| 13 | app_transactions                    | 0.038808   |
| 10 | sale_product_views                  | 0.037876   |
| 12 | product_detail_view_per_app_session | 0.037599   |
| 11 | discount_rate_per_visited_products  | 0.036907   |
| 6  | total_product_detail_views          | 0.035277   |
| 0  | account_length                      | 0.033566   |
| 1  | user_id                             | 0.033544   |
| 8  | promotion_clicks                    | 0.031643   |
| 2  | add_to_wishlist                     | 0.025854   |
| 17 | push_status_yes                     | 0.021952   |
| 18 | location_code_415                   | 0.006539   |
| 19 | location_code_510                   | 0.005455   |

The most influential factors driving customer behavior are **desktop\_sessions**, **customer\_service\_calls**, and **app\_sessions**, suggesting that both online engagement and customer support interactions strongly affect outcomes.

Lower importance for **location codes** and **push notifications** indicates that geography and alerts play a minor role in predicting customer actions.

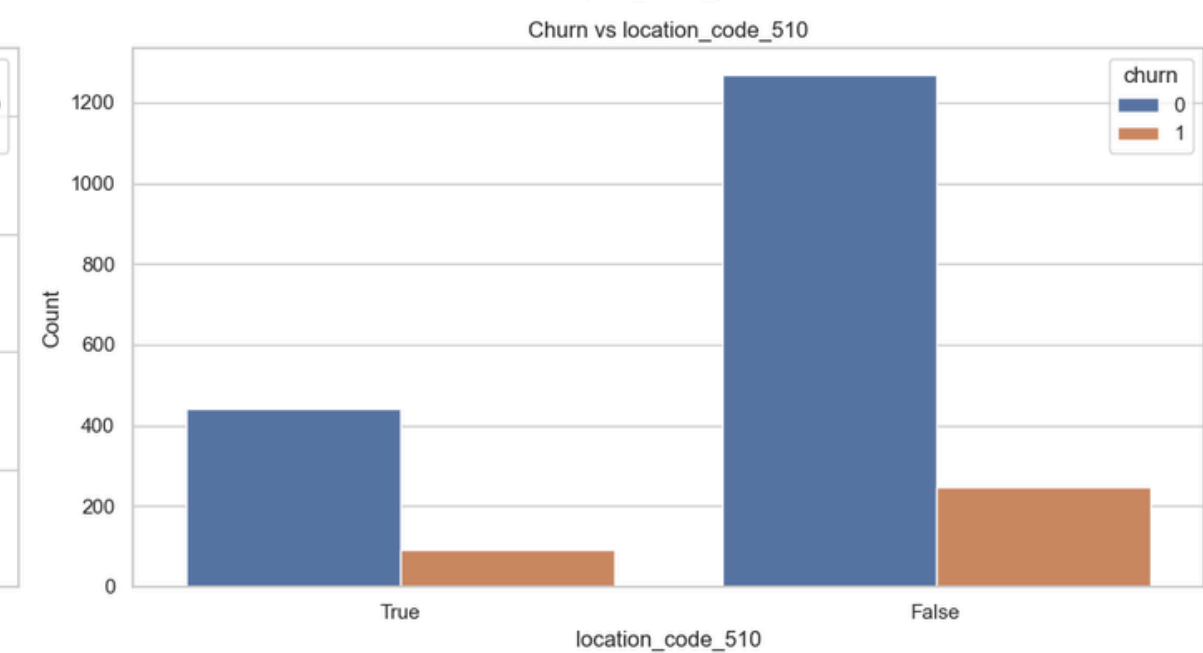
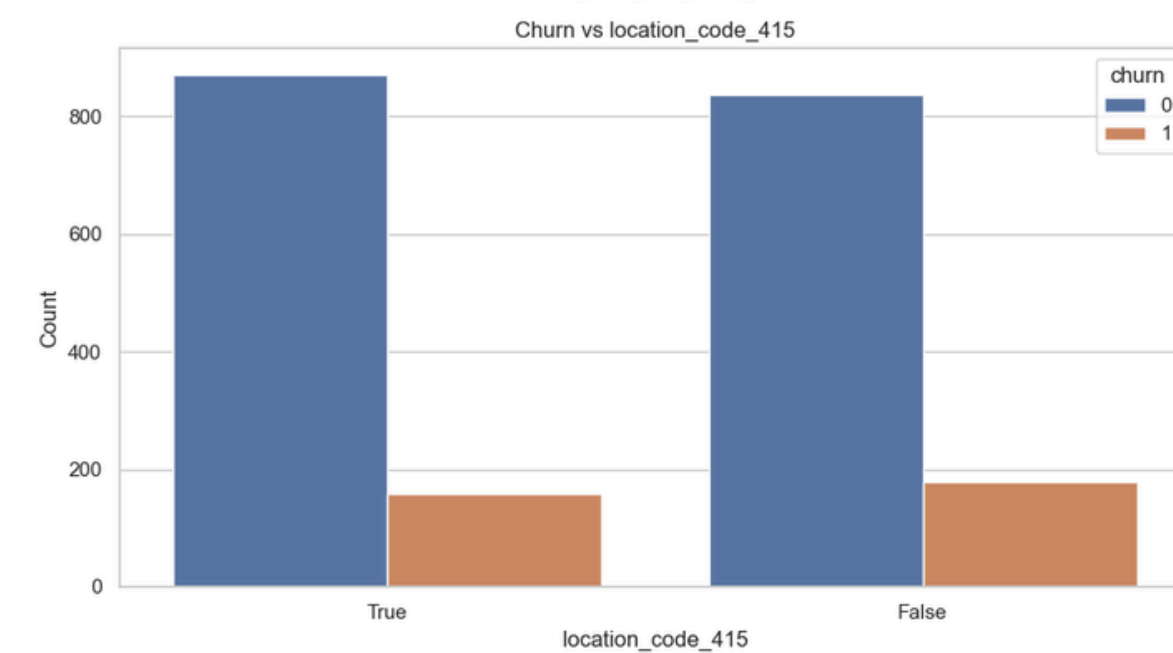
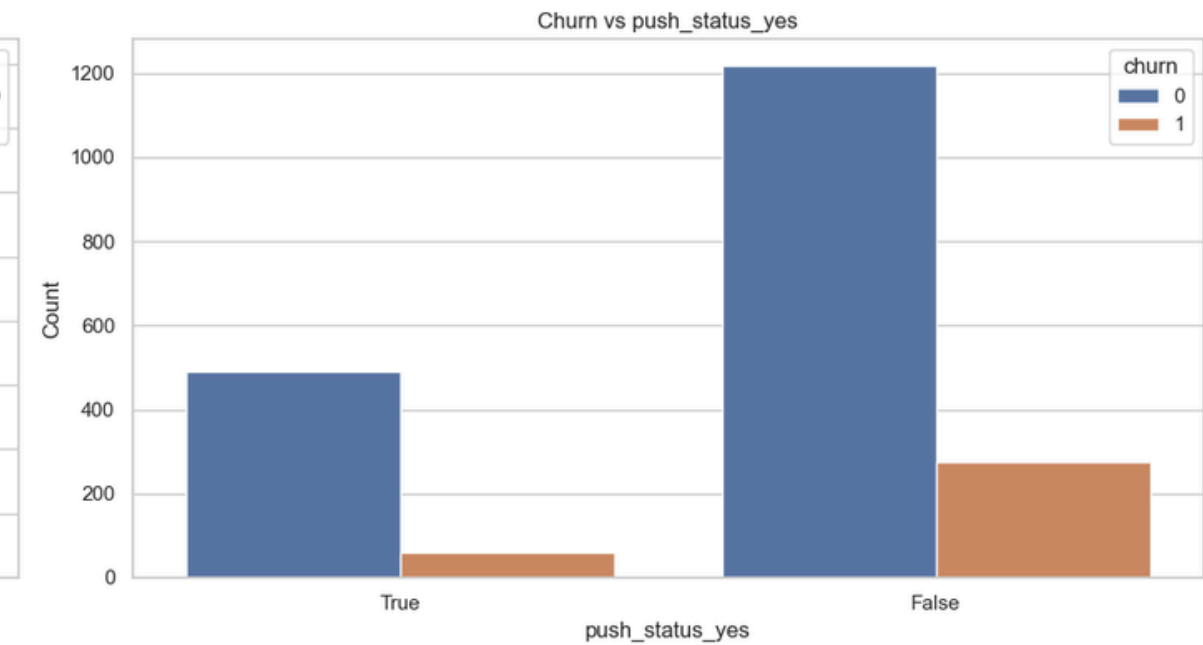
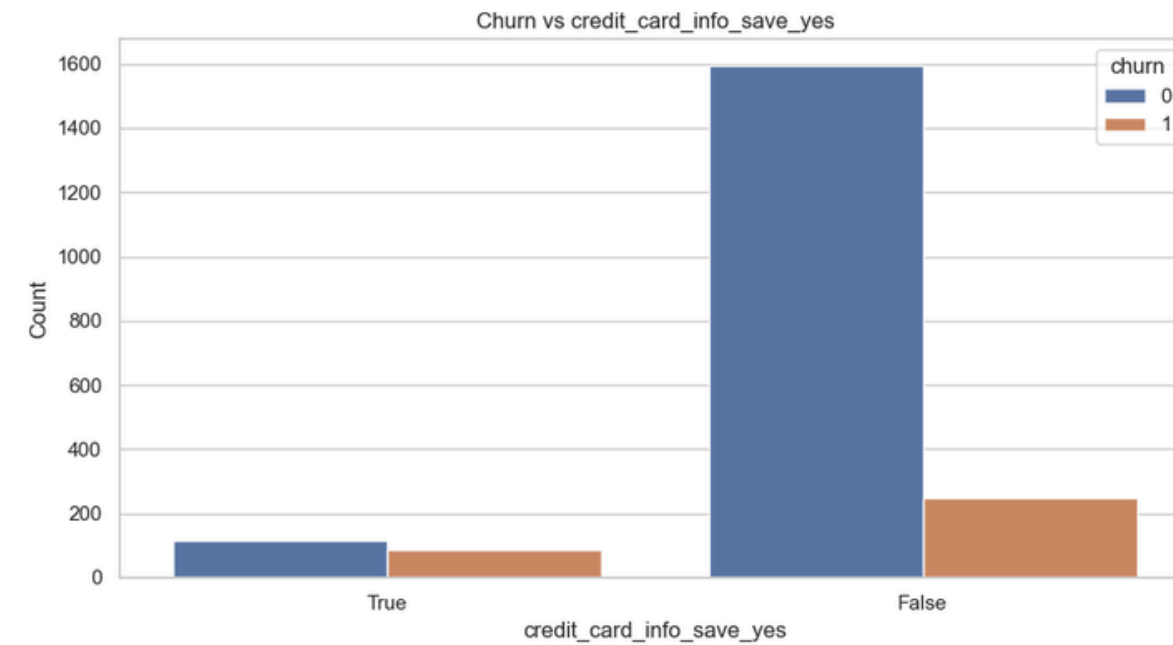
# OBSERVATION FROM THE CATEGORICAL DATA

Categorical Data – push\_status, credit\_card\_info\_save, location\_code



- Majority of customers are in location code 415, with a significantly higher count compared to Location Code 510.
- Majority of customers don't allow push notifications, as indicated by the larger number of users in the "False" category for push status.
- Majority of customers don't store their credit card information on the side, as the "False" category for credit card info save far exceeds the "True" category.

# CHURN AGAINST THE CATEGORICAL DATA



## Insights:

- Credit Card Info Save:** Customers who save their credit card info are much less likely to churn. Not saving info correlates with higher churn.
- Push Status:** Customers with push notifications enabled have lower churn rates. Most customers without push notifications churn more.
- Code 415:** This location has a lower churn rate compared to others, indicating better customer retention.
- Code 510:** Churn is slightly higher here than in Location Code 415, suggesting the need for more targeted retention efforts.

## Recommendations:

- Encourage saving payment info to reduce churn.
- Promote push notifications to engage and retain customers.
- Target Location Code 510 with specific retention strategies.
- Maintain efforts in Location Code 415 to keep churn low.

The background features two large, light pink geometric shapes. One is a triangle on the left side, and the other is a larger, more complex shape on the right side, both pointing towards the center.

**THANK YOU!**